SSU Data Dash 2026: Toronto Home-Price Prediction

TORONTO FOURCAST | Yan Cui · Jiayi Guan · Shengde Wang · Ziang Chen

Five-minute video: [pending recording] Code and predictions: public GitHub repository

Summary

We trained separate gradient-boosted models for the off-market and list-aware questions. Across six monthly walk-forward validations from December 2025 through May 2026, the models reached MdAPE of **10.60%** off-market and **4.36%** list-aware on 10,069 homes per track. These scores are below the starter benchmarks of 20.8% and 7.3%, respectively. Adding community, municipality, and house style reduced MdAPE in all six months; remaining off-market errors are largest for the lowest- and highest-priced homes.

Data and features

We used the provided 150,000 synthetic sales from January 2021 through May 2026. Numeric features are square-footage midpoint, bedrooms, washrooms, parking, latitude, longitude, and months since January 2021. We added property type, house style, municipality, and community as categorical features. Off-market never uses list price; list-aware adds it. Garage and lot dimensions were tested but showed no reliable paired-validation gain. We excluded listing remarks because the organizers have not clarified whether they contain price signal beyond the structured fields; third-party AVM estimates are unavailable for the test sets. We used **no outside data**.

Models and validation

Both tracks use scikit-learn's histogram gradient-boosting regressor trained on log sale price. It handles categories natively. Each fold learns category values from that fold's training sales; unseen categories become missing values. The list-aware model receives list price because it is available at prediction time. The off-market model does not.

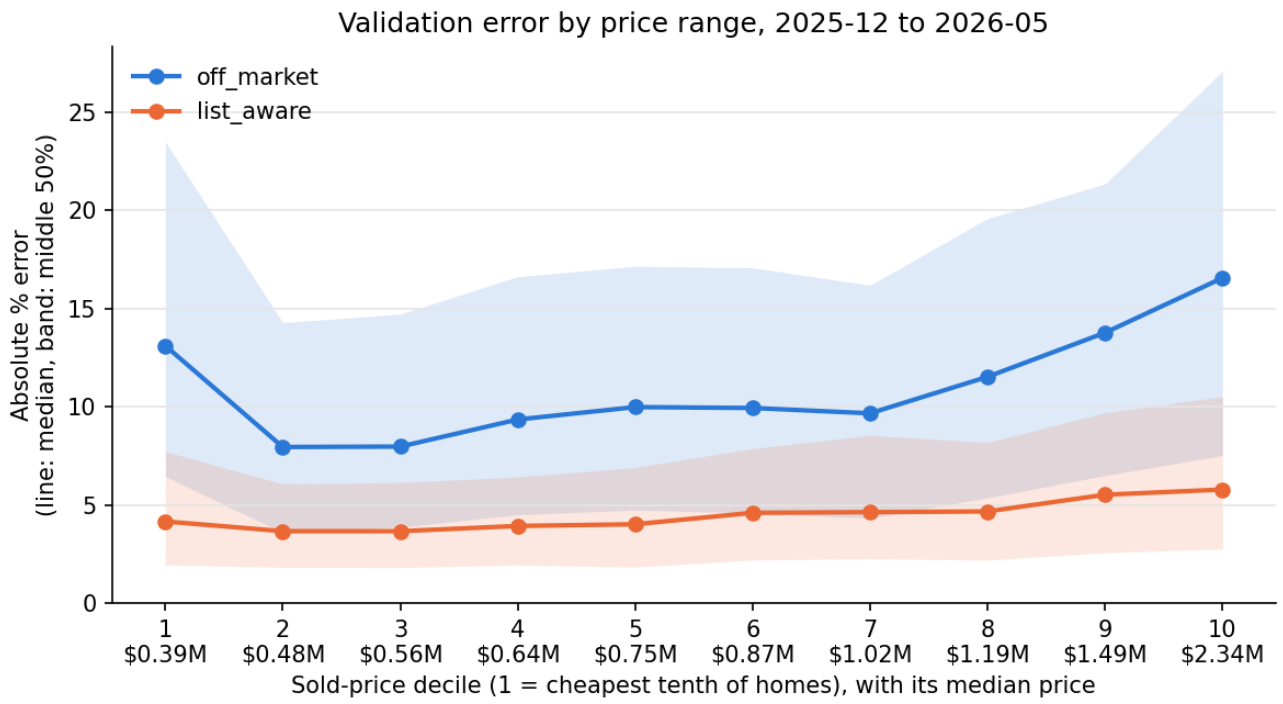
For each month from December 2025 to May 2026, the model trained on sales before that month and predicted that month's sales. We did not use hidden test prices. The intervals below bootstrap individual homes and do not account for errors that share the same month's market conditions.

Results

Track	Starter MdAPE / MAPE	Current MdAPE (95% CI)	Current MAPE (95% CI)	MdAPE change
Off-market	20.8% / 27.4%	10.60% (10.38-10.85)	13.40% (13.17-13.65)	-10.2 pp
List-aware	7.3% / 8.1%	4.36% (4.26-4.46)	5.57% (5.47-5.66)	-2.9 pp

Compared with our numeric and property-type feature model on the same homes, adding the three categories reduced MdAPE by 3.80 percentage points off-market (paired 95% bootstrap interval: 3.52 to 4.12) and 0.19 points list-aware (0.14 to 0.24). All six months improved on both tracks.

Where errors remain



Absolute percentage error by shared sold-price decile. The line is the median; the shaded band is the middle 50%.

Off-market MdAPE is 13.10% in the cheapest decile and 16.55% in the most expensive, compared with about 8% to 12% through the middle. List-aware MdAPE rises to 5.79% in the top decile from 3.67% in the second and third deciles. Off-market Co-op Apt has 14.33% MdAPE, but that estimate uses only 48 validation homes. These segments describe the synthetic validation set and do not establish why individual homes were mispriced.

Limitations and outside data

Validation covers six months and may differ from the hidden June-September 2026 distribution. Home-level bootstrap intervals understate uncertainty if errors correlate within months. The dataset is synthetic: community names and coordinates do not map to real Toronto locations, so these results are not a live-market valuation service. A next step is to test segment-level calibration for the price tails using the same forward-only validation. **Outside data used: none.**

Reproducing the predictions

From a clean clone: `python -m pip install -r requirements.txt`, then run `python run.py validate`, `python run.py report`, `python run.py predict`, and `python run.py check`. The check command verifies the two required CSVs. Add the recorded, tested YouTube link before submitting.